

Handheld Real-Time ECG MEASUREMENT SYSTEM

DR GEETALI SAHA AND KUSHAL GADHVI

An electrocardiogram (ECG) acts as the health indicator of our heart. With the help

of electrodes, placed either on the chest or at the arms, we can sense the beating of the heart. The signals

can be visualised as waveforms on a computer screen or printed on a sheet.

ECG is a graphical representation of the variation in electric potential of the heart during one cardiac cycle. Fig. 1 shows the authors' prototype of the handheld real-time ECG measurement system and Fig. 2

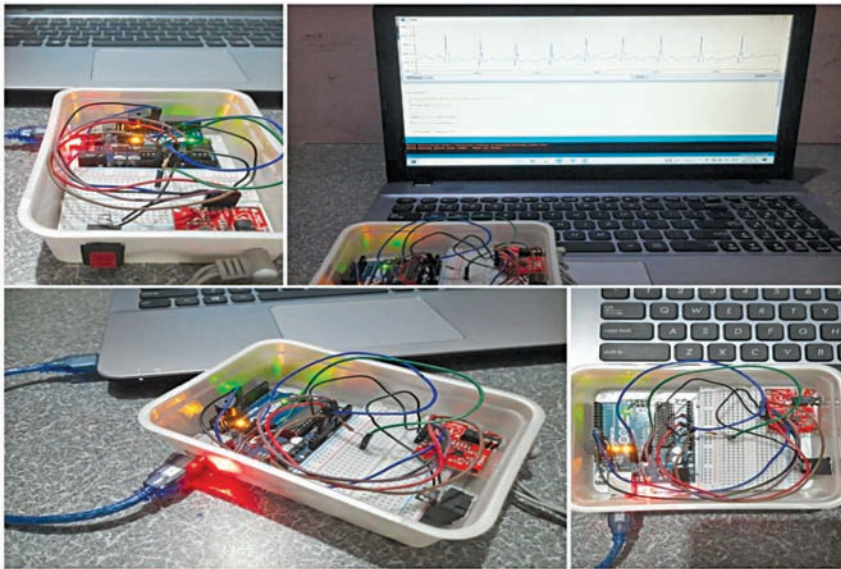


Fig. 1: Authors' prototype of the handheld real-time ECG measurement system

TABLE 2
BOARD LABELS AND PIN FUNCTIONS OF ARDUINO UNO

Board label	Pin function	Arduino Uno connection
GND	Ground	GND
3.3V	3.3V power input	3V3
Output	Analogue output of the sensor	A0
LO-	Leads-off Detect-	D8
LO+	Leads-off Detect+	D9
SDN	Shutdown	Optional

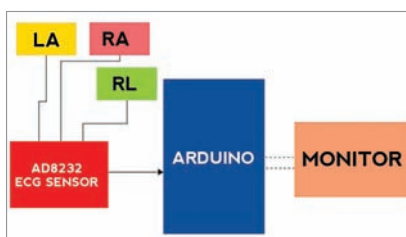


Fig. 2: Block diagram of the handheld real-time ECG measurement system

TABLE 1
BILL OF MATERIAL

Components	Description	Quantity
AD8232 sensor (MOD1)	ECG sensor module	1
Electrodes	3M ECG adhesive type electrode	3
Arduino Uno (MOD2)	Microcontroller	1
Buzzer (B1)	3-5V buzzer	1

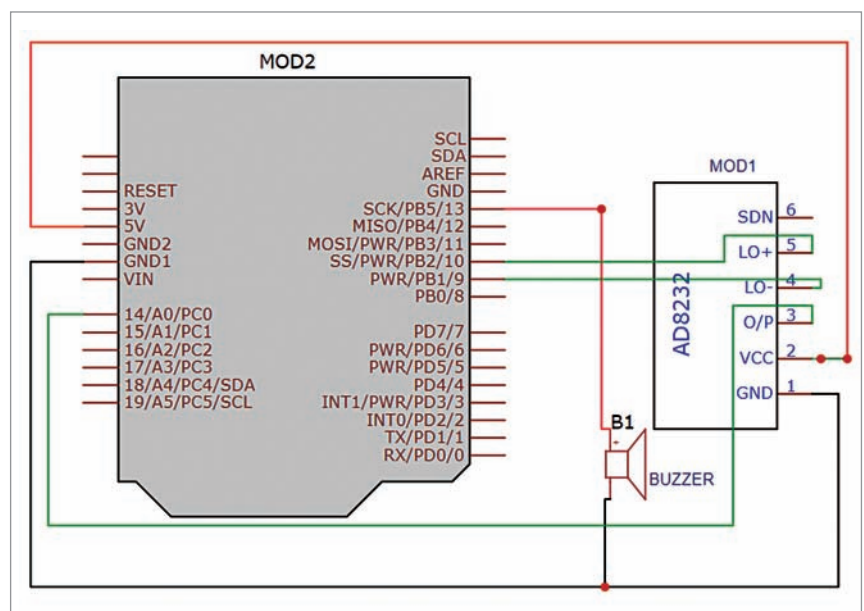


Fig. 3: Circuit diagram of the system